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Positive-Sequence Phasor Modeling of Droop-Controlled, Grid-Forming Inverters with Fault Current Limiting Function

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***Abstract*—Traditional positive-sequence phasor models of droop-controlled, grid-forming inverters do not have the fault current limiting function. During a short-circuit fault the models generate unrealistic high fault currents making the simulation results less practical. This paper develops a fault current limiting function for the positive-sequence phasor model of droop-controlled, grid-forming inverters, which can effectively limit the inverter output current at the predefined maximum during faults. A user-written model has been developed for the commercially available software Siemens/PTI PSS/E. Fault studies on a modified IEEE 39-bus system with all grid-forming inverters verify the effectiveness of the developed fault current limiting function. The proposed model can be used to evaluate how the limited fault currents of droop-controlled, grid-forming inverters impact the bulk power system transient stability under fault conditions.**

***Index Terms*—fault current limiting, droop control, grid-forming inverter, modeling, phasor, positive-sequence**

I. Introduction

The massive increase of inverter-based resources (IBRs) in bulk power systems has caused concerns in utilities all over the world due to the reduction of system inertia and strength. To address this issue, researchers in this area are actively working on designing new controls for IBRs that aim to improve the stability of inverter-based power systems [1].

The grid-forming inverter is considered as one of the key technologies to improve the stability of power systems with high penetration of IBRs [2]. Compared to traditional grid-following inverters, the grid-forming inverters directly control the inverter output voltage and frequency so that they can work without synchronous generators. Therefore, it is possible to realize a 100% inverter-based power system using grid-forming inverters [1]. In recent years, numerous studies have shown that grid-forming inverters can significantly improve the frequency stability of power systems [3-5]. The work in [3] compared the performance of grid-forming and grid-following inverters, and the study results show that grid-forming inverters can improve both the system frequency stability and small signal stability. The case study in [4] showed that the grid-forming photovoltaic inverters with sufficient headroom can improve the frequency stability of the Hawaiian island of Oahu. The work in [5] indicated that the transition from 0% to 100% inverter-based transmission systems is achievable from the frequency stability point of view.

Although it is becoming clear that grid-forming inverters with sufficient headroom can improve the system frequency stability, one major issue of grid-forming inverters that has not been fully addressed is the fault current limiting control and its potential impacts on the system transient stability. Because the grid-forming inverter approximately behaves as a voltage source behind an impedance, it provides very high current during faults which could damage the inverter hardware that typically only tolerates 1.5-2.5 pu overcurrent [6, 7]. Different fault current limiting controls have been proposed for grid-forming inverters [8-10]. The virtual impedance method has been used to limit the over-currents of grid-forming inverters [8, 9]. However, while this method works for balanced faults, it could have issues with unbalanced faults. The work in [10] designed a fault current limiting control that works for both balanced and unbalanced faults. Although these fault current limiting controls have been verified by laboratory experimental testing, the inverter models developed are mostly electromagnetic transient models, which are not suitable for the large-scale system simulation because of the high computational burden.

To evaluate the impacts of limited fault currents of grid-forming inverters on the transient stability of large-scale transmission systems, the positive-sequence phasor model of grid-forming inverters with fault current limiting function should be developed. The work in [4] and [11] developed the positive-sequence phasor and three-phase phasor models of droop-controlled, grid-forming inverters for the dynamic simulation of large-scale transmission and distribution systems,

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respectively. The work in [12] developed positive-sequence phasor models for both virtual oscillator control and droop control, and compared their performances in a modified IEEE 39-bus test system and the microWECC test system. The work in [13] developed a dynamic phasor model for grid-forming inverters and analyzed their responses to faults. These phasor models can be used to evaluate the system dynamic stability under relatively small disturbances, such as the loss of generators. However, because of the lack of fault current limiting function, they are not suitable for evaluating the system transient stability under severe faults.

This paper develops a fault current limiting function for the positive-sequence phasor model of droop-controlled, grid-forming inverters, which can effectively limit the inverter output current at the predefined maximum during faults. A user-written model has been developed for Siemens/PTI PSS/E [14]. Fault studies on a modified IEEE 39-bus system with all grid-forming inverters verify the effectiveness of the developed fault current limiting function. The developed model can be used to evaluate how the limited fault currents of droop-controlled, grid-forming inverters impact the bulk power system transient stability.

II. Grid-Forming Inverter Concept

Being different from grid-following inverters that behave as controllable current sources, a grid-forming inverter behaves as a controllable voltage source behind a coupling reactance as shown in Fig. 1. The internal voltage magnitude E and angular frequency ω are controlled by the inverter controller.

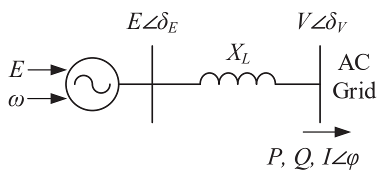

Fig. 1 Basic model of grid-forming inverter.

The coupling reactance, X_L, is important for the controller design. By properly sizing X_L, for example, between 0.05 and 0.15 pu on an inverter rating base, the inverter output active power, P, and reactive power, Q, can be approximately decoupled. As shown by (1) to (3), P is approximately linear with the phase angle difference δ_P, and Q is approximately linear with E. The well-developed droop control is based on this decoupling characteristic [15].

$$\delta_p = \delta_E - \delta_V \tag{1}$$

$$P = \frac{EV}{X_L}\sin\delta_p \approx \frac{EV}{X_L}\delta_p \tag{2}$$

$$Q = \frac{E^2 - EV\cos\delta_p}{X_L} \approx \frac{E(E-V)}{X_L} \tag{3}$$

III. Positive-Sequence Phasor Modeling of Droop-Controlled, Grid-Forming Inverters

This section will introduce the developed positive-sequence phasor model of droop controlled, grid-forming inverters, including the well-developed droop control, the fault current limiting function, and how the model interacts with the transmission network solver.

A. Droop Control Modeling

Fig. 2 (a) and (b) show the well-developed droop control, including the Q-V droop control, the P-f droop control, and the overload mitigation control. Extensive tests have been conducted at the Consortium for Electric Reliability Technology Solutions (CERTS) Microgrid testbed in the past decades, verifying the effectiveness of this control strategy [16, 17].

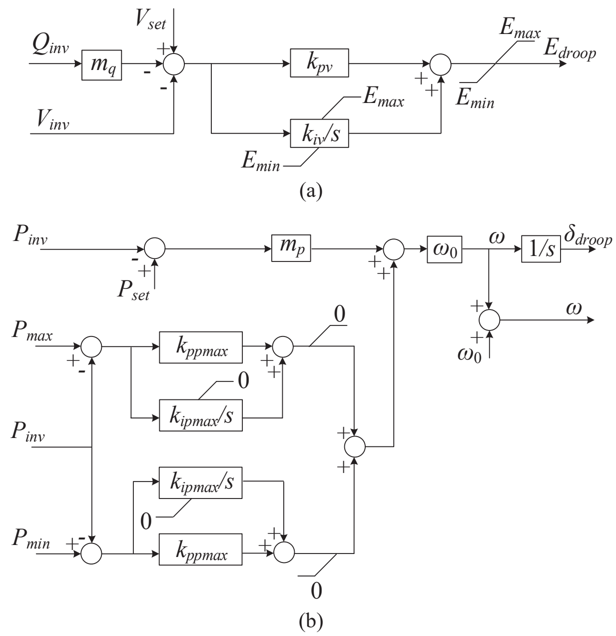

Fig. 2 Droop control. (a) Q-V droop control. (b) P-f droop control and overload mitigation control.

Because the controller has already been well explained in previous references [16, 17], this paper will not focus on explaining its physical meaning. Table I lists all the inverter parameters. When interacting with the transmission network solver, the per unit values of P, Q, V on the system rating base returned by the network solver need to be converted to the per unit values on the inverter rating base and pass through a low-pass filter as shown by (4) to (6), where S_{base} is the base value of the system rating, M_{base} is the base value of the inverter rating, and P_{inv}, Q_{inv}, and V_{inv} are the per unit values of inverter output active power, reactive power, and voltage magnitude on the inverter rating base. The outputs of the controller, E_{droop} and δ_{droop}, are used to determine the inverter internal voltage $E\angle\delta_E$.

$$P_{inv} = \frac{1}{1+T_f s} P \frac{S_{base}}{M_{base}} \tag{4}$$

$$Q_{inv} = \frac{1}{1+T_f s} Q \frac{S_{base}}{M_{base}} \tag{5}$$

$$V_{inv} = \frac{1}{1+T_f s} V \tag{6}$$

TABLE I. PARAMETERS OF THE DROOP-CONTROLLED, GRID-FORMING INVERTERS ON THE INVERTER RATING BASE

Symbol	Description	Value
X_L	Inverter coupling reactance	0.15 pu
m_q	Q-V droop gain	0.05 pu
V_{set}	Voltage set point	Initialized by power flow
k_{pv}	Proportional gain of the voltage controller	0 pu
k_{iv}	Integral gain of the voltage controller	5.86 pu/s
E_{max}	Upper limit of the output of the voltage loop	1.2 pu
E_{min}	Lower limit of the output of the voltage loop	0 pu
m_p	P-f droop gain	0.01 pu
P_{set}	Power set point	Initialized by power flow
P_{max}	Upper limit of the inverter output power	1 pu/s
P_{min}	Lower limit of the inverter output power	0 pu/s
k_{ppmax}	Proportional gain of the overload mitigation controller	0.01 pu
k_{ipmax}	Integral gain of the overload mitigation controller	0.1 pu/s
ω_0	Rated angular frequency	376.99 rad/s
T_f	Time constant of the low-pass filter for P, Q, and V measurements	0.01 s
I_{max}	Inverter maximum output current	2 pu

B. Fault Current Limiting Function Modeling

One major issue for the droop control shown in Fig. 2 is that it will force the inverter to provide very high current during faults because of the voltage source behavior. However, according to the testing results reported by [6] and [7], the commercially available grid-forming inverters typically can only provide 1.5 pu to 2.5 pu fault currents. Therefore, it would not be practical to use the model shown in Fig. 2 to evaluate the transient stability of power systems because of the unrealistic high fault currents provided by inverters.

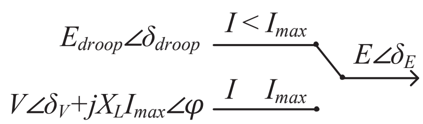

Fig. 3 Fault current limiting function.

This paper develops a fault current limiting function for the droop-controlled, grid-forming inverters based on the model shown in Fig. 2. Fig. 3 shows the developed fault current limiting function. The inverter works in the droop control mode during normal operations and keeps monitoring its output current $I\angle\varphi$. When the magnitude of the output current I is smaller than the maximum current limit I_{max}, the inverter internal voltage is governed by the droop control so that $E\angle\delta_E = E_{droop}\angle\delta_{droop}$. However, once I exceeds I_{max} because of severe faults, the inverter internal voltage $E\angle\delta_E$ will be calculated based on the inverter terminal voltage $V\angle\delta_V$, the coupling reactance X_L, and the new current phasor $I_{max}\angle\varphi$ as shown by Fig. 3. By doing so the magnitude of the inverter output current I will be limited at I_{max} during faults, but its phase angle φ will remain unchanged compared to the case without the fault current limiting function. Once the fault is cleared, the inverter output current will drop below I_{max} so that the operation mode will autonomously switch back to the droop control mode.

It is important to note that this paper focuses on developing a positive-sequence phasor model that can represent the limited fault current behavior of droop-controlled, grid-forming inverters, while it does not focus on designing the inverter control strategy to limit the fault currents. The experimental testing results in [6] and [7] indicate that the output currents of grid-forming inverters during faults are limited by directly manipulating the inverter pulse-width modulation (PWM) signals. Such high-frequency switching control cannot be modeled in the phasor domain. Therefore, this paper develops a fault current limiting function that can effectively represent the limited fault current behavior of droop-controlled, grid-forming inverters, but it does not show how the actual control is implemented at the inverter PWM control layer.

C. Interface with the Transmission Network Solver

When interfacing with the transmission network solver, the inverter internal voltage source needs to be converted to its Norton equivalent circuit for the network solution, as shown in Fig. 4. Equation (7) and (8) are used for the conversion.

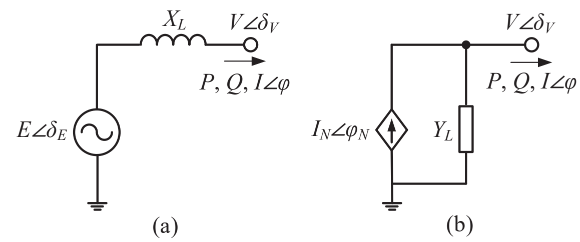

Fig. 4. Inverter equivalent circuits. (a) Inverter internal voltage source and the coupling reactance. (b) Norton equivalent current source.

$$I_N\angle\varphi_N = \frac{E\angle\delta_E}{jX_L} \tag{7}$$

$$Y_L = \frac{1}{X_L} \tag{8}$$

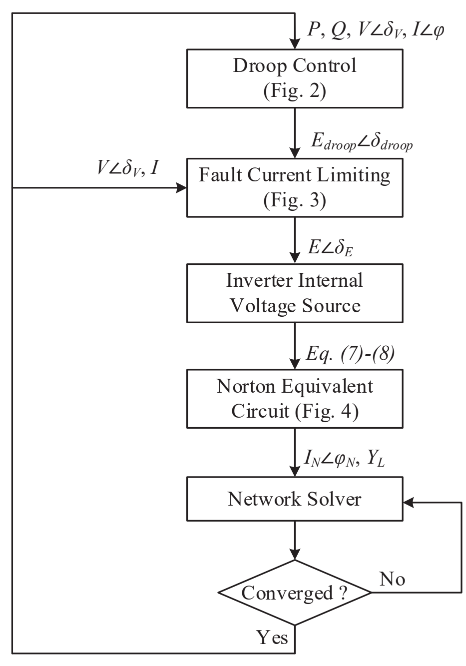

Fig. 5 Interaction between the grid-forming inverter model and the transmission network solver.

The flowchart in Fig. 5 shows the process of how the developed positive-sequence phasor model interacts with the transmission network solver.

IV. Simulation

A user-written model of the droop-controlled, grid-forming inverters with the fault current limiting function has been developed for the positive-sequence phasor simulator Siemens/PTI PSS/E. Simulations were performed on the IEEE 39-bus system with all the 10 synchronous generators replaced by 10 equivalent droop-controlled, grid-forming inverters, which form a 100% inverter-based transmission system. The 10 grid-forming inverters have the same parameters on the inverter rating base as listed in Table 1. Specifically, they have a 1% *P-f* droop, a 5% *Q-V* droop, and a maximum current limit I_{max} of 2 pu. A bolted fault is applied to Bus 19 at 5 s and cleared after 0.1 s. Two scenarios are simulated to examine the developed fault current limiting function and evaluate the system transient stability. The simulation time step is 4.2 ms.

A. Scenario 1: Fault Current Limiting Function Disabled

Fig. 6–Fig. 8 show the responses of the 10 droop-controlled, grid-forming inverters to the fault when the fault current limiting function is disabled.

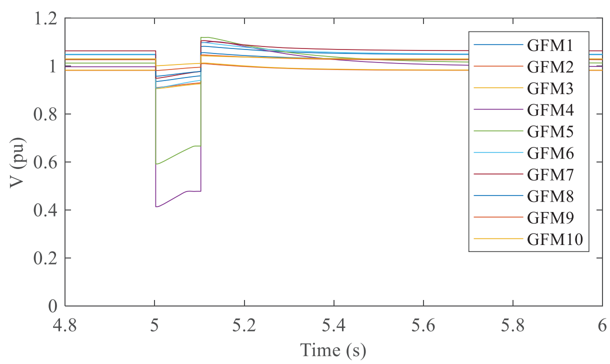

Fig. 6 Output voltages of grid-forming inverters when the fault current limiting function is disabled.

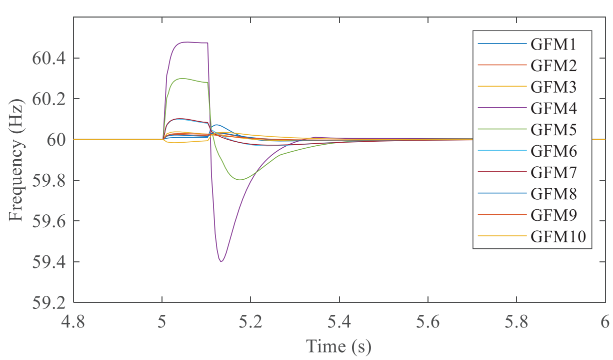

Fig. 7 Frequencies of grid-forming inverters when the fault current limiting function is disabled.

Although the voltage and frequency responses shown in Fig. 6 and Fig. 7 indicate that the system can maintain stability after the fault is cleared, Fig. 8 shows that the output currents of grid-forming inverter 4 and 5 (GFM4 and GFM5) exceed the maximum 2 pu because they are close to the fault location. These significant over-currents result in unrealistic simulation results and not accurately reflect the transient stability conditions.

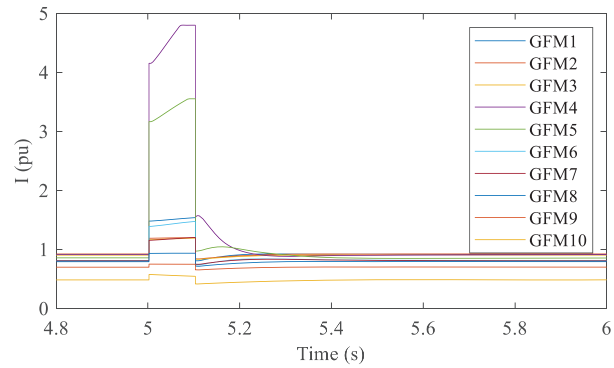

Fig. 8 Current magnitudes of grid-forming inverters when the fault current limiting function is disabled.

B. Scenario 2: Fault Current Limiting Function Enabled

Fig. 9–Fig. 12 show the responses of the 10 droop-controlled, grid-forming inverters to the fault when the fault current limiting function is enabled.

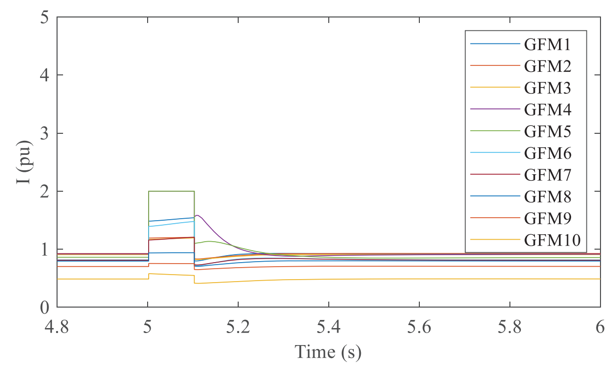

Fig. 9 Current magnitudes of grid-forming inverters when the fault current limiting function is enabled.

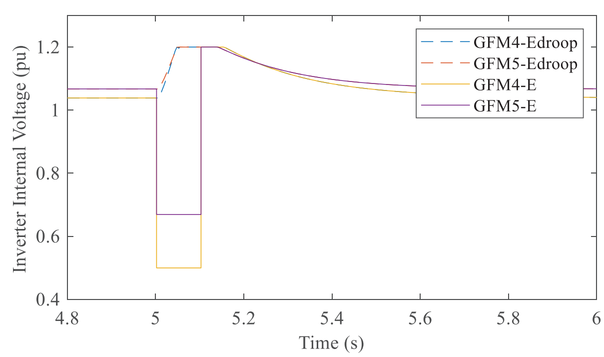

Fig. 10 Internal voltages of GFM4 and GFM5 when the fault current limiting function is enabled.

As shown in Fig. 9, the developed fault current limiting function can effectively limit the fault currents of GFM 4 and GFM 5 at the maximum 2 pu. This is because once the inverter output current exceeds I_{max}, the inverter internal voltage E will switch from E_{droop} to the new voltage generated by the fault current limiting function to limit the output current, as explained in Section III B. Fig. 10 shows E_{droop} and E of GFM4 and GFM5. Although E_{droop} remains high during the fault, the actual internal voltage E is significantly reduced by the fault current limiting function, resulting in a limited output current.

Once the fault is cleared, the output current I drops below I_{max} so that E autonomously switches back to E_{droop}.

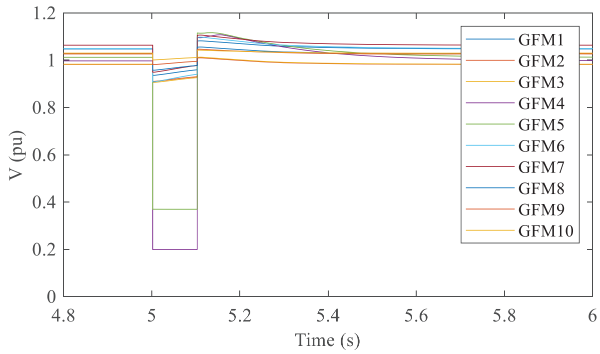

Fig. 11 Output voltages of grid-forming inverters when the fault current limiting function is enabled.

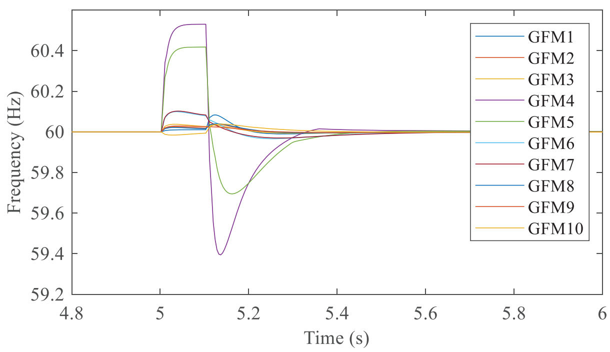

Fig. 12 Frequencies of grid-forming inverters when the fault current limiting function is enabled.

Fig. 11 and Fig. 12 show the voltage and frequency responses of the 10 grid-forming inverters when the fault current limiting function is enabled. By comparing Fig. 6 and Fig. 11 it can be seen that during the fault the output voltages of GFM4 and GFM5 are lower when the fault current limiting function is enabled. This is because the limited output currents from inverters result in the reduction of the output voltages. By comparing Fig. 7 and Fig. 12 it can be seen that the frequency transients are not significantly affected by the fault current limiting function.

V. Conclusion

Traditional positive-sequence phasor models of droop-controlled, grid-forming inverters do not have the fault current limiting function. During a short-circuit fault the models generate very high fault currents resulting in unrealistic simulation results. This paper has developed a fault current limiting function for the positive-sequence phasor model of droop-controlled, grid-forming inverters, which can effectively limit the inverter output current at the predefined maximum during faults. A user-written model has been developed for the commercially available software Siemens/PTI PSS/E. Fault studies on a modified IEEE 39-bus system with all grid-forming inverters have verified the effectiveness of the developed fault current limiting function for the positive-sequence phasor model. The model can be used to evaluate how the limited fault currents of droop-controlled, grid-forming inverters impact the transient stability of bulk power systems under severe faults.

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